3.5 The Chain Rule_Guide

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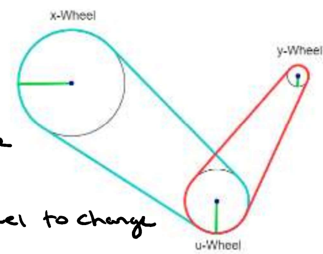
Motivation:
We wish to determine how to take the derivative of a composition of functions in multiple dimensions, i.e., what does chain rule look like for functions beyond $f(g(x))$?

Exploration:
Consider the mechanism to the right. Suppose that every time the x-wheel spins one time, the u-wheel spins 2 times. Furthermore, every time the u-wheel spins 1 time, the y-wheel spins 3 times. Question: If the x-wheel spins one time, how many times does the y-wheel spin? [1]



The y wheel will spin 6 times for every 1 time the x-wheel spins

If we look at the ratio of change in the u-wheel to change in the x-wheel we get $\frac{du}{dx} = \frac{2}{1}$

Similarly $\frac{dy}{du} = \frac{3}{1}$

So $\frac{du}{dx} \cdot \frac{dy}{du} = 2 \cdot 3 = 6 = \frac{dy}{dx}$

Recall: In calculus one you learned that if $y = f(x)$ and $x = g(t)$ where f and g are differentiable functions, then y is indirectly a differentiable function of t and $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$.

Note: We will see in this section that the chain rule generalizes nicely to functions of several variables.

> **Theorem (Case 1 of the Chain Rule):**
> Suppose that $z = f(x, y)$ is a differentiable function of x and y where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t. Then z is a differentiable function of t and
> $$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

Proof: See pages 949 – 950 & Appendix D of the text.

Note: One may see the above equation written as: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$.

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[1] The corresponding applet can be found here:
https://webspace.ship.edu/msrenault/geogebracalculus/derivative_intuitive_chain_rule.html

Example 1: Suppose that $z = \cos(x+4y)$, $x = 5t^4$, and $y = \frac{1}{t}$. Assuming $t \neq 0$, find $\frac{dz}{dt}$.

$4t^{-1}$

Approach 1: $z = \cos\left(5t^4 + \frac{4}{t}\right)$

$$\frac{dz}{dt} = -\sin\left(5t^4+\frac{4}{t}\right)\cdot\left(20t^3 - \frac{4}{t^2}\right)$$

SAME

Approach 2:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial z}{\partial y}\cdot\frac{dy}{dt}$$

$$= \left(-\sin(x+4y)\right)\cdot\left(20t^3\right) + \left(-4\sin(x+4y)\right)\left(-\frac{1}{t^2}\right)$$

Example 2: The pressure P (in kilopascals), volume V (in liters), and temperature T (in kelvins) of a mole of an ideal gas are related by $P = \frac{8.31T}{V}$. Find the rate at which the pressure is changing (rounded to the nearest thousandth) when the temperature is 300K and increasing at a rate of 0.1 K/s and the volume is 100 L and increasing at a rate of 0.2 L/s.

Note P is a multivariable function $P(T,V)$ & T & V depend on time t.

We want $\frac{dP}{dt}$

$\frac{2}{V}$

$$\boxed{\frac{dP}{dt} = \frac{\partial P}{\partial T}\cdot\frac{dT}{dt} + \frac{\partial P}{\partial V}\cdot\frac{dV}{dt}}$$

Always the same

$$= \left(\frac{8.31}{V}\right)\cdot(0.1) + \left(-\frac{8.31T}{V^2}\right)\cdot(0.2)$$

$\frac{dP}{dt}$ when $V = 100$ & $T = 300$ is $\left(\frac{8.31}{100}\right)(0.1) + \left(-\frac{8.31(300)}{100^2}\right)(0.2)$

So $\frac{dP}{dt} \approx -0.042$ kaP/sec

> **Theorem (Case 2 of the Chain Rule):**
> Suppose that $z = f(x,y)$ is a differentiable function of x and y, where $x = g(s,t)$ and $y = h(s,t)$ are differentiable functions of s and t. Then, $\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$ and $\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$.

__Example 3:__ Suppose that $z = y(2^x)$ where $y = \overset{4}{s^2 + st}$ and $x = \overset{1}{e^{st}}$. Find the exact value of $\frac{\partial z}{\partial s}$ when $\underline{s = 2}$ and $\underline{t = 0}$.

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial s} = \left(y\cdot 2^x\cdot \ln(2)\right)\cdot\left(e^{st}\cdot t\right) + \left(2^x\right)\left(2s+t\right)$$

$$\left.\frac{\partial z}{\partial s}\right|_{\substack{s=2\\t=0}} = \overset{0}{\cancel{\left(4\cdot 2\cdot \ln(2)\right)\left(e^0\cdot 0\right)}} + (2)\cdot(4+0) = 8$$

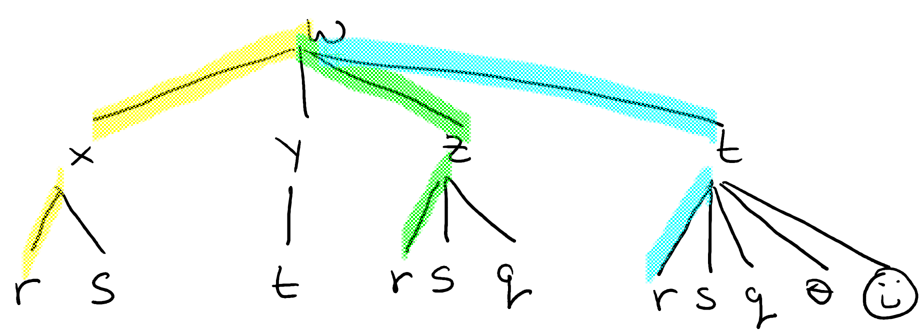
Note: It is important to see that the symbol ∂z *does not have a meaning in and of itself (unlike* dz*). Therefore, one should recognize that* $\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \neq \frac{\partial z}{\partial s} + \frac{\partial z}{\partial s}$ *(unless it is the case that* $\frac{\partial z}{\partial s} = 0$*).*

> **Theorem (General Chain Rule):**
> Suppose that u is a differentiable function of the n variables $x_1, x_2, \ldots, x_n$ and each x_j is a differentiable function of the m variables $t_1, t_2, \ldots, t_m$ and $\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1}\frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2}\frac{\partial x_2}{\partial t_i} + \cdots + \frac{\partial u}{\partial x_n}\frac{\partial x_n}{\partial t_i}$ for each $i = 1, 2, \ldots, m$.

Proof: The proof the case 2 of the chain rule and the general chain rule are beyond the scope of the course.

Visualizing the General Chain Rule:
A great way to think through the general chain rule is with the help of a tree diagram. To make the idea of the tree diagram concrete, I will use a simple example.

$w = f(x,y,z,t)$ & $x = x(r,s)$ & $y = y(t)$, $z(r,s,q)$ & $t = t(r,s,q,\theta,☺)$



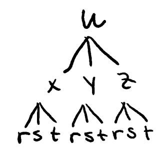
Suppose you want to find $\frac{\partial w}{\partial \text{☺}}$

then $\frac{\partial w}{\partial \text{☺}} = \frac{\partial w}{\partial t} \cdot \frac{\partial t}{\partial \text{☺}}$

Suppose you want to find $\frac{\partial w}{\partial r}$

then $\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial r} + \frac{\partial w}{\partial t} \cdot \frac{\partial t}{\partial r}$

Example 4: Suppose that $u = x^4 y + y^2 z^3$, $x = \overset{2}{r} s e^t$, $y = \overset{2}{r} s^2 e^{-t}$, and $z = r^2 s \sin(\overset{0}{t})$. Find the exact value of $\frac{\partial u}{\partial s}$ when $r = 2$, $s = 1$, and $t = 0$.



$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial s}$$

$$= (4x^3 y)(re^t) + (x^4 + 2yz^3) \cdot (2rse^{-t}) + (3z^2 y^2)(r^2 \sin(t))$$

then evaluate at $r = 2$, $s = 1$ & $t = 0$

$$\frac{\partial u}{\partial s} = (64)(2) + (16 + 0)(4) + (0)(0)$$

$$= 128 + 64$$

$$= 192$$